

SS 2 GEOGRAPHY WEEKS 1-6

WEEK ONE

Topic: Earth's external process (action of running water)

Action of running water (Rivers)

Running water is one of the most important agents of denudation. Rivers are involved in erosion, transportation and deposition of materials.

Terms Associated with Rivers are:

- 1) Source of a River: The source of a river refers to where a river starts or begins, usually around highlands.
- 2) Course of a River: This refers to the path or channel through which the river flows.
- 3) Mouth of a River: This refers to where the river ends or where it enters into the sea, ocean or lake.
- 4) River basin or Catchment Area: This refers to all areas drained by a river and its tributaries.
- 5) Watershed or Water divide: This refers to the highland area which separates two or more rivers or two river basins. It is from the watershed that rivers take their sources.
- 6) River regime: This refers to the seasonal changes in the volume of water in a river in a year. It could be single regime, where there is one period of high volume and one period of low volume, and the double regime where there are two distinct periods of high volume in a year.

Knowledge of a river regime is important to man in controlling possible floods, storing up water for irrigation and human consumption, and for planning H.E.P production (Hydro-electric power).

- 7) Confluence of a River: This refers to the meeting point of two different rivers.
- 8) Tributaries: These are smaller rivers or streams that join together to form a larger one.
- 9) Distributaries: These are channels formed by the division of a river as it flows into the sea. They are usually found in the delta region of a river.

10) River energy: This refers to the velocity of a river. The efficiency of a river to erode and transport the eroded materials depends very much on its velocity in erosion, transportation and deposition of materials.

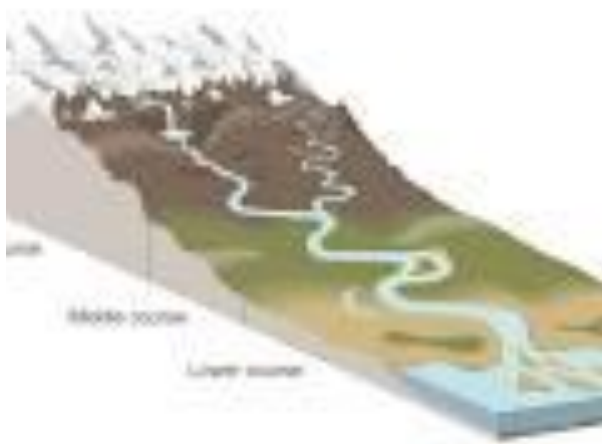
Factors Affecting the Velocity of a River

- a) The volume of water released: The higher the volume of water released by a river, the higher the velocity. However, as the volume decreases during the dry season, the velocity also decreases.
- b) Slope of the river valley: The steeper the slope, the higher the velocity of the water in the river.
- c) Shape of the river valley: A river uses more energy to flow through a flat-wide valley than through a narrow-deep valley because the former has a large surface area.
- d) Amount and size of materials: The greater the speed of a river, the greater the materials or loads it can carry or move. A river uses its energy to (a) overcome friction along its channels, (b) to transport its load and (c) to erode the river valley. It is this energy that is trapped by man to generate electricity (H.E.P).

Stages of a River

The entire length, valley or course of a river is divided into three main parts or stages. These are:

- i) The upper or mountain course (youthful stage)
- ii) The middle or valley course (mature stage)
- iii) The lower or plain course (old age stage)



[Diagram: Fig 1.1 Stages of a River —

The Upper Course of a River — Characteristics

- a) It marks the beginning or source of a river
- b) It is found around highland areas
- c) It has steep sides
- d) The dominant work is vertical corrasion or erosion

Processes of River Erosion

The load or materials carried by a river are the main agents of erosion, but the erosive works of a river consists of four interacting processes. These are:

- a) Hydraulic Action: In this process, fast flowing water forces itself into cracks and cavities within the valley under pressure and enlarges the cracks.
- b) Corrasion: Corrasion is the wearing away of the sides and floor of the river valley with the aid of sand, pebbles, silts, and boulders which are being transported.
- c) Attrition: This is the wearing down of the load as they collide with one another and with the floor and sides of the valley. Large boulders are broken down into small pieces like pebbles.
- d) Solution: This refers to the chemical action of water on materials it comes in contact with while flowing. Here, rocks and salts are dissolved and carried away in solution.

Features of Upper Course of a River

The main features in the upper course of a river include:

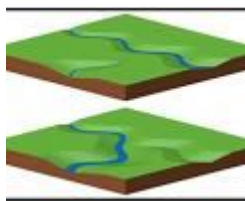
- a) V-Shaped Valley: This is formed as a result of vertical corrasion or erosion of the river. The river valley is deepened and narrowed into a steep-sided V-shaped valley. The valley is widened by weathering and mass wasting which cause it to become V-shaped.
- b) Gorge: This is also formed due to deepening and narrowing of the river as a result of vertical erosion. As a result, the valley is deep and narrow with vertical sides, resulting in the formation of a gorge. A gorge may develop if a river follows a line of weakness such as a fault. A formation of large and impressive gorge is called a Canyon.
- c) River Capture: This is also called river piracy or river beheading. A river capture refers to two parallel consequent rivers with tributaries separated by a more powerful

river and with presence of wind gap, e.g.s are river Ogunpa (Ibadan) in Nigeria, river Irrawady in India.

Conditions for River Capture

River capture occurs when a river through headward erosion flows into a water divide and cuts off the head water of another weaker river. For river capture to occur, the following conditions must be available:

- i) There must be two parallel consequent rivers separated by a divide or have a common watershed.
- ii) One of the rivers must be more powerful or have more erosive power than the other.
- iii) The valley of the more powerful river must be deeper than the other.
- iv) It must cut back into the headland (headward erosion) faster than the other river.



[Diagram: Fig 2.1 River Capture

d) Rapids and Cataracts: These features are formed due to uneven resistance of hard and soft rocks which are crossed by a river. A minor upstream or forward jump of such water in a river is called a rapid, while a jump of greater dimension is called a Cataract.

e) Waterfall: This is formed when a bar of resistant rock lies across a river valley. It may also result when a fault line scarp lies across a river valley, or develops from a rapid, or when water plunges down a plateau edge. It has a steep slope which forms plunge pools. Examples are Kwa falls, Victoria falls, Niagara falls, Erin-Ijesa falls, Gurara falls, etc. Waterfalls are important for generation of hydro electric power and tourism (tourist centre).

Processes of River Transportation

The load of a river is carried or transported along the course of a river through four main processes. These are:

- a) Solution: Some rock materials dissolve in water and are carried in solution from the upper to lower course of the river.
- b) Suspension: Here, the lighter particles of solid materials are carried in suspension as the water flows.
- c) Saltation: In this process larger particles are moved in series of hops or jumps along the stream bed.
- d) Traction: In this process very large fragments of materials are rolled or pushed along in the river.

Importance of Rivers to Man

- 1) Rivers serve as a medium of transportation from one place to another.
- 2) They also help in the generation of Hydro-electric power (H.E.P) where rapids and waterfalls exist.
- 3) Rivers also provide water for irrigation purposes.
- 4) Rivers also provide water for domestic and industrial purposes.
- 5) Formation of flood plain by rivers also provides fertile soil for agricultural activities.
- 6) Rivers also provide food e.g. fish, prawns, crabs etc.
- 7) Rivers provide employment to many people e.g. fishermen and canoe builders.
- 8) Some rivers act as political boundaries between states, regions and nations.

Assignment: 1 Briefly describe 4 features of wind erosion
2. List 2 features of wind deposition in desert.

Week: Two

Topic: Action of wind

Action of Winds in Deserts

The action of wind is dominant in deserts and other arid regions of the world. Deserts are arid regions of the world which are associated with (aridity) and (dryness) caused by low rainfall, high temperature, cold current and high evaporation.

Examples of Deserts

Sahara desert (West Africa) — the largest desert

Kalahari and Namib Desert (South Africa)

Arabian, Iranian and Thar Desert (Middle East)

Australian desert (Australia)

Atacama Desert (South America)

Mohave Desert (U.S.A.)

Characteristics of Deserts

1. Extremes of temperature: deserts do experience extreme high temperature during the day and extreme low temperature at night.
2. Low Rainfall: Deserts do experience low rainfall due to cold currents blowing them.
3. Absence of Vegetation Cover: There is lack of dense vegetation covers in deserts because of low rainfall.
4. High evaporation rate: As a result of high temperature, there is high rate of evaporation.
5. Wind action is dominant: As a result of the above characteristics, the action of wind tends to be dominant in deserts.
6. Presence of Cold Currents: Cold currents are known to be responsible for the cause of deserts.

Types of Deserts

There are five distinct kinds of deserts landscape. These are:

1. Erg or Sandy deserts: these are made up of sand and sand dunes or sheets.

2. Hamada or rocky deserts: these are made up of mainly rocks.
3. Reg or Stony deserts: these are made up of mainly pebbles and gravel.
4. Badlands: these are desert highly eroded by wind.
5. Mountain deserts: these deserts are associated with highlands areas.

In deserts, mechanical (exfoliation) and chemical weathering are more active in breaking rocks to sand than wind erosion.

Features of Wind Erosion in the Desert

Rock Pedestals

Rock pedestals are irregular in shape e.g. mushroom shape with alternate horizontal layers of hard and soft rocks (like pillars in structure), being narrow at the base and broader at the top. Rock pedestals are formed by wind abrasion on an alternate horizontal layers of hard and soft rocks.

Zeugen

These are tabular masses with a layer of soft rocks lying beneath a surface layer of hard rocks. Zeugen is formed when a tabular mass of rock which has a layer of soft rocks lying under a layer of a more resistant hard rock is lying horizontally to the direction of wind. The mass of rock is then attacked by wind abrasion.

Yardangs

Are formed when hard and soft rocks in vertical bands are aligned in the direction of prevailing winds. These hard rocks are called Yardangs.



Fig 3.1 Yardang

Mesas and Buttes

Is a flat table-like landmass. It has a resistant horizontal top layer with steep sides and made up of soft and hard layers. As a result of the action of denudation, the hard top layers of rock resists agents of denudation and protects the softer layers of rocks below from being eroded.



Fig 3.2 Mesa and Butter

Inselberg

Is an arid land form. It is an isolated rocky outcrop having steep sides, round top and composed of granite. Inselberg is formed from the existence of extensive old plateau. It is caused as a result of weathering and removal of weathered materials by water and wind. Inselberg serves as tourist centres (tourism). It's also a site for quarrying...

Deflation Hollows or Depressions

They are formed by wind deflation. It is a desert landform with varying depths. It's very extensive and can form oasis in deserts.

Features of Wind Deposition in the Deserts

1. Dunes

Dunes are hills or ridges of sand formed by the piling up of sand onto hill-shaped by the action of wind. Dunes could be active or inactive with various shapes. There are two main types of dunes. These are:

a) Barchan

Is a crescentic or moon shaped structure. They may occur in group or singly. A barchan has a convex shape on windward side and a concave shape on the leeward side. It is formed by wind deposition in deserts. They are desert landforms.

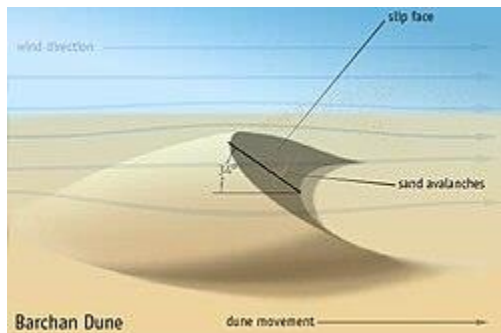


Fig 3.3 **Barchan**

b) Seifs or Longitudinal Dunes

Seifs are sword shaped. They have long and narrow ridges of sand usually of hundreds of kilometres. They are formed by wind deposition in deserts.



Fig 2.1 Seif or Longitudinal Dune

2. Loess

These are loose, fine particles which are carried by wind. They are deposited outside the desert as loess. Loess is a fine loam, very fertile and porous. Deposit of loess is found in North West China.

Assignment: 1 Define glacier

2.mention 2 features of glacier erosion

Week: Three

Topic: Glacial Actions

Action of Glaciers in Temperate Regions

The action of glacier as an important agent of erosion, transportation and deposition of materials is related to mountainous and temperate regions of the world.

Terms Associated with Glaciers

Ice: This refers to solid form of water, resulting from freezing when the temperature is persistently below 0°.

Glacier: Meaning moving ice, refers to a large accumulation of ice (frozen sea or river) in motion.

Glaciation: This refers to the wearing away of the earth surface by glacier.

Snow: This refers to frozen water vapour which falls in form of crystals through the atmosphere.

Snow-Line: This is the lower limit of perpetual snow cover on a mountain.

Snow field: This refers to the region or area permanently under snow cover.

Features of Glacier Erosion in the Highland Areas

1. **Striations:** These are scratches or marks left on rocks over which glacier passes. Rock fragments or mountains embedded in the glacier affect the striation of rocks.
2. **Corries or Cirque:** A corrie is a deep and rounded hollow or depression with steep sides formed through erosion by ice. They are arm-chair shaped hollow resulting from the plucking of rock materials down the slope. After the ice has moved, it later forms corrie lake.
3. **Arete:** This occurs when two corries cut back opposite sides of the same mountain, resulting in a knife edged ridge called Arete. Arete is therefore a wall-ridge-like structure separating two corries.
4. **Pyramidal peak:** This occurs where three or more corries cut back on the same mountain. A pinnacle shaped like a pyramid, develops and is called Pyramidal peak

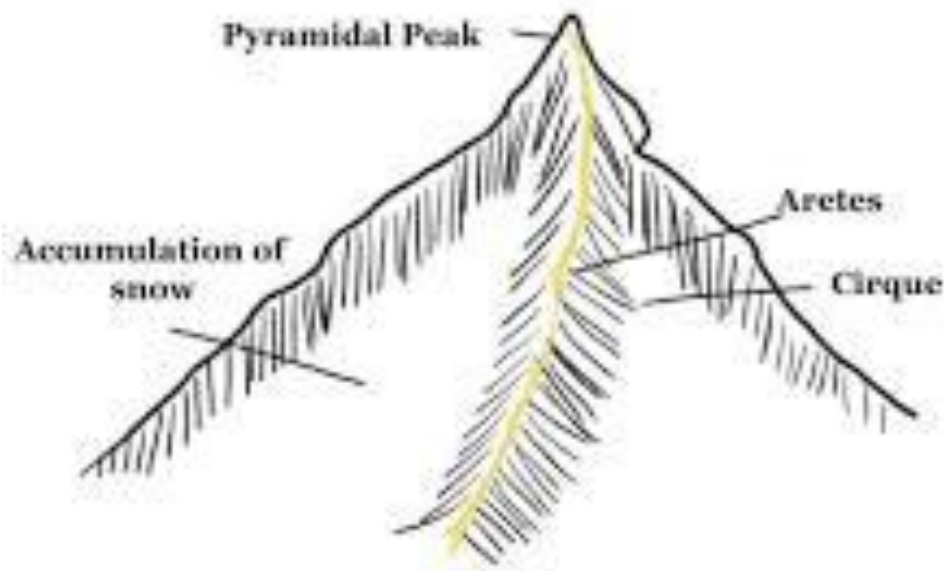


Fig 11:2 — Cirque, Arete and Pyramidal peak (diagram showing Pyramidal peak,

Bergschrund: This is formed at or near the head of glacier. It is deep and vertical crack separating the lower part of a glacier that has started to move down its valley and the upper part which is static

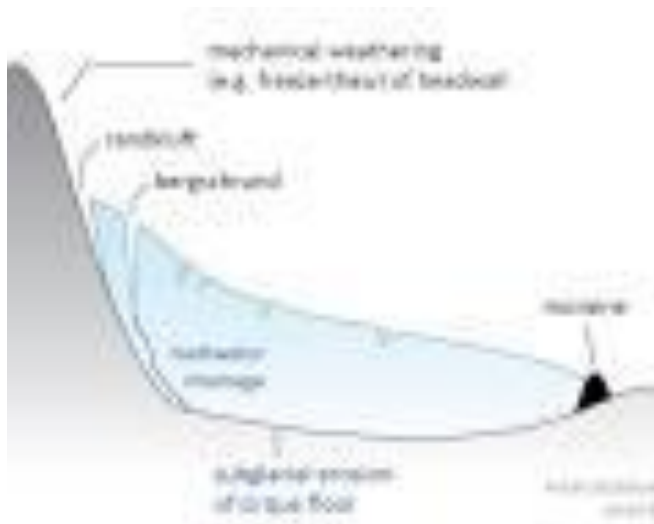


Fig 11:3 — Bergschrund

6. **U-shaped Valley or Trough:** This is a wide, flat floor with very steep side which has been eroded by glacier. All the sides and floor, including all debris are washed away by glacier.

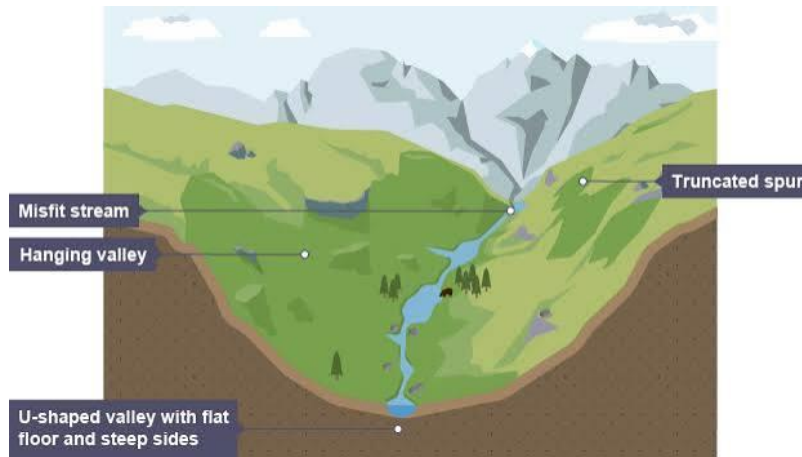


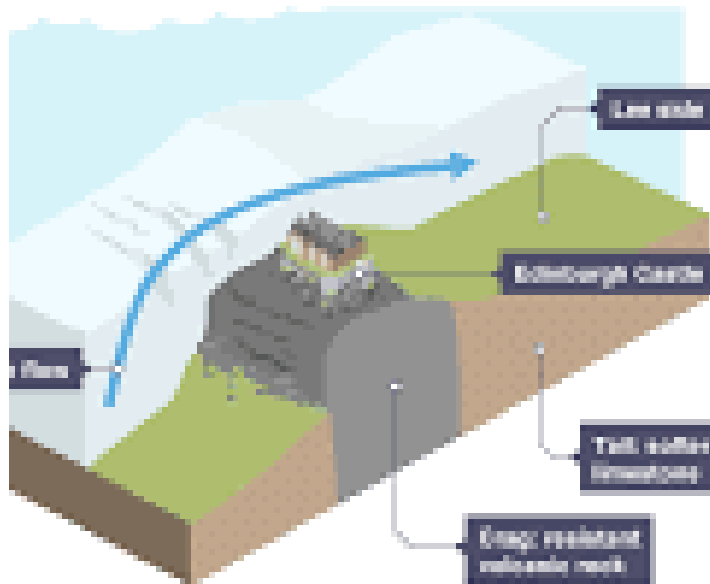
Fig. 11.4 Glaciated highland

7. **Hanging Valley:** This results when glacier action from tributaries erodes materials into the main valley. It can serve to generate electricity.
8. **Rock Basin and Rock Step:** When two tributaries join a main valley, the additional weight of ice in the main valley cuts deeper into the valley floor at the point of convergence, thus forming a rock step. Rock basin is formed on the other hand when glacier erodes and excavates in the valley floor so deep to form the rock basin.
9. **Moraines:** are made up of pieces of rocks that are shattered by frost action, embedded in the glacier and borne down the valley. Moraines exist in Lateral, medial, ground and in terminal forms.

Glacial erosion in Lowland

Erosional features of glacier which occurs in Lowland. Areas include:

1. **Roche Moutonne:** These are resistant residual rock structures. The surface is striated by ice movement. It's upstream side is smooth due to abrasion while the downstream is rough due to plucking
2. **Crag and tail:** crag is a mass of hard rock which slope on the upstream side that protects the softer leeward slope from erosion and later develop to form the tail.



Features of Glacier Deposition in Lowlands

1. Boulder clay
2. Erratic's
3. Drumlins
4. Eskers
5. Terminal moraines
6. Outwash plains

Economic Importance of Glacier

In some semi-arid areas glaciers are of considerable economic importance in the irrigation of crops.

In mountainous areas such as Switzerland, Norway, and Iceland, meltwater from glaciers is critical to the economic generation of hydroelectric power.

Their landscape value: Tourism. This is especially true when visiting glacierized areas like the Scottish highlands and Yosemite National Park in California.

Assignment:

1. What is wave
2. Mention 3 mechanism of wave erosion

Week: Four

Topic:: Action of waves

Action of Waves in Coastlands

The action of waves as an important agent of erosion, transportation and deposition of materials is restricted to the coast of seas and oceans. The rate of marine erosion depends on the nature of the rocks, the amount of rock exposed to the sea, the effects of tides and currents and human interference in coast protection.

Terms Associated with Waves

Waves: Wave means the turbulent movement of water as a result of wind moving over the water.

Tides: Tide is the alternate rise and fall of the surface of the sea and the sea approximately two times a day.

Current: Ocean Current is the movement of the surface water of the ocean in a definite direction.

Coast: This refers to the meeting point between the land and the sea.

Shore: This is the part of the land that lies between high water and low water.

Beach: This refers to the materials deposited on the shore by the action of waves.

Swash: This refers to the water thrown up the beach by breaking waves.

Backwash: This is the water that sucks back and retreats after wash.

Undertow: This is the water which flows near the bottom away from the shore.

Mechanism of Wave Erosion

1) Corrosion: This is the wearing down of the base of the cliff by wave action.

2) Attrition: This is the breaking down of materials like pebbles, boulders, etc. against each other, as the wave continues its activities into smaller particles.

3) Hydraulic Action: In this process, fast moving waves force themselves into cracks and cavities within the base of the cliff under pressure and enlarge the cracks.

4) Solvent action: This involves the disintegration of rock materials such as limestone in the coast by chemical action of the sea.

Features Produced by Wave Erosion

1) Cape and Bays

They are features characteristics of marine erosion in Coastline which can be made of hard rock (Cape) or soft rock (bays). Cape and bay in other words are formed due to the action of waves on rocks of varying resistance which causes the coast to be eroded irregularly.



CAPE AND BAY

2) Cliffs

These are steep rock faces adjoining the coast. They are formed due to the action of waves on the base of the headland which cuts backwards as the wave action intensifies.



Fig 11.27 — Cliff (sketch: "Cliff" and "Sea")

3) Coastal Cave

Is formed by wave erosion when the softer rocks are more quickly eroded than the harder rocks. The pressure in the cave increases by the compression of the air, and the results in the expansion of the hollow, which forms a cave.

4) Arch

When two caves approach each other from either side of a headland, they meet and unite to form an arch.

5) Stack

It is formed by wave erosion. There is a presence of headland. Wave erosion attacks headland both sides.

6) Stump

When the stack is seriously eroded to a point that a small portion is just visible above the sea level, a stump is said to be formed.

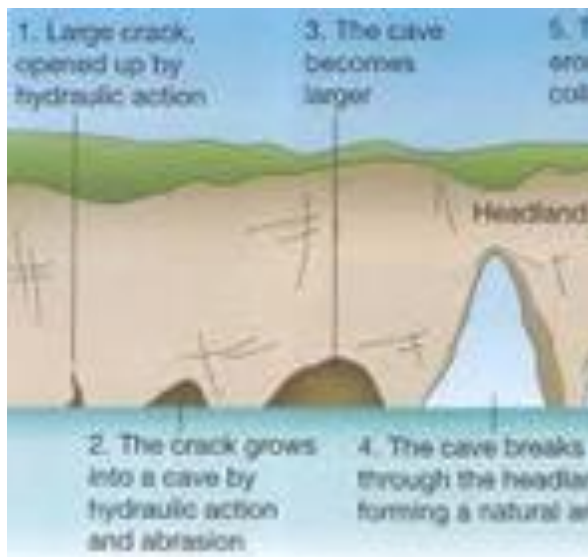


Fig 11.28 — Cave, Arch, Stack and Stump (sketch with labels: Cave, Arch, Stack, stumps)

7) Geo

Geo develops when a wave cuts into a cliff, resulting in a narrow hole called geo. It is formed when a cave collapses



Fig 11.29 — Geo (sketch: "Cliff," "Geo," "Sea")

8) Gloop or Blowhole

Is formed by wave erosion on a Coastal Cliff. There is less resistant rock or line of weakness at the base of the cliff. Cave is produced. The spray water in the cave enlarges a vertical joint in the roof while the hydraulic action of the waves causes loose blocks to fall in while waves breaking into the cave may force water or air out of the hole.



GLOUP

Features of Coastal Deposition

1) Beaches

Beaches are made up of sand and gravel. They are depositional features on the coast. Beaches are formed when sand and gravel loosened from the land are moved by waves to be deposited while sifted along the shore.

2) Spits

These are ridges of sand and gravel formed by longshore drift across the entrance to an estuary, lying on one side to the land and the other into the ocean.

3) Sand Bar

Are formed by wave deposition in shallow sandy waters.

4) Marine Dunes and Dune belts:

Dunes are on shore wind with large force which make a large amount of coastal sand to move landward to form marine dune.

Assignment;

- 1 what is climate
- 2 State 5 factors affecting Climate

Week: five

Topic: climate

Meaning of Climate

Climate is defined as the average weather condition of a place over a long period of time usually taken between 30 and 40 years. The weather of any given place changes from day to day while climate is comparatively constant over a long time.

Elements of Climate

The elements of climate include: temperature, rainfall, wind, relative humidity, cloud cover, pressure and sunshine.

1) Temperature:

Heat within the air, indicates how hot or cold a place is, and varies from latitude to latitude. The sun is the source of all heat, measured with the aid of thermometer.

2) Pressure:

This consists of a number of mixed gases and has weight which it exerts on the earth's surface. It varies from place to place and from time to time. Pressure is measured with a barometer.

3) Cloud:

These are particles of water droplets in the upper atmosphere suspended in the air due to condensation of water vapour carried by rising air. Clouds may give precipitation.

Types of cloud include Cirrus, Cumulus, and Stratus.

4) Sunshine:

This is the visible part of the solar energy. It is useful for plants and other living organisms. The amount of sunshine depends on the latitude of a place, and the slope of the land. Sunshine varies according to season.

5) Rainfall:

This refers to water droplets from the atmosphere. It occurs as a result of evaporation from water bodies e.g. rivers, lakes and oceans, and transpiration from plants. The amount of rainfall is measured with the aid of a rain gauge. The different types of rainfall are Convectional, Orographic and Cyclonic.

6) Wind:

This is air in motion, from high pressure to low pressure belts. It has a special mode of direction and is made up of a series of dusts and eddies, capable of carrying moisture, dust and other pollutants. The direction is measured with a wind vane while the speed is measured with an anemometer.

7) Humidity:

This is a measure of the dampness of the atmosphere due to water in the gaseous state and it varies from place to place at different times of the day.

Factors Affecting Climate

Climate varies from one part of the world to another due to the effects of the following factors:

1) Latitude:

Latitude refers to the location of a place in relation to the equator. Tropical latitudes (where the sun's altitude is always high) are hotter in temperature than latitudes where the sun's altitude is generally low.

2) Altitude:

Altitude refers to the height of a place above sea level. As one moves higher into the atmosphere, the temperature decreases by 6.5°C for every 1000m of ascent. This is called the Normal Lapse Rate.

3) Continentality (Distance from Sea):

Generally, the further inland a place is from the sea, the lower the rainfall it receives; the closer to the sea, the higher the rainfall.

4) Ocean Currents:

Ocean currents can affect the climate of adjacent coastland. Warm currents raise the temperature of adjacent coastland; cold currents help the formation of fogs along the coast adjacent to land.

5) Planetary Winds and Pressure Belts:

Winds can affect the climate of adjacent coastal belts. For example, warm winds raise the temperature of adjacent coastlands, while cold winds lower the temperature of adjacent coastlands. Warm, moist winds from the ocean bring rainfall to adjacent coastlands, e.g. the West Monsoon in West Africa.

6) Slope and Aspect:

A steep slope experiences more rapid change in temperature than a gentle slope.

LAPSE RATE

LAPSE rate is defined as the rate of change in temperature with altitude or the decrease in air temperature as one ascends into the atmosphere. A regular air temperature decrease with an average of 0.65°C for every 100metres of ascent. This rate is called the lapse rate.

LAPSE rate is described as 0.65°C per 100m of ascent or 6.5°C per 1000m of ascent.

Example 1

If the temperature at the foot of a mountain is 29°C , what will be the temperature at its summit (top) if the mountain is 4000m High?

Solution:

LAPSE rate= 6.5°C per 1000m of ascent.

Therefore, 4000m of ascent= $6.5^{\circ}\times 4000\text{m}\div 1000\text{m}$
 $=6.5^{\circ}\times 4=26^{\circ}\text{C}$

Importance of Climate to Man

1) Human Settlement:

Weather and climate affect the rate at which human beings live in a place. Temperate areas are more habitable than desert areas because of differences in climate.

2) Health:

A humid or damp environment encourages the growth of micro-organisms which cause diseases of man; hence the death rate is higher in tropical than in temperate countries.

3) Environmental Hazards:

Hazards like soil erosion, rainstorm, flood, and drought are caused by differences in weather and climate of a place.

4) Vegetation:

Temperature and rainfall are the major determinants of the type of vegetation in an area, which could be thick or light.

5) Clothing:

The type and nature of clothes worn by different people in different parts of the world are due to differences in climate. Cold climate requires black and thick dresses while hot climate requires light dresses.

6) Housing:

The type and nature of houses built in different places are also influenced by the differences in climate. For example, hot regions require air-conditioned houses whereas polar regions do not.

7) Agriculture:

Rainfall and temperature are important elements that determine the type of crops to be cultivated in an area.

8) Soil Formation:

Soil is formed from parent rocks which have been broken down by the elements of weather and climate.

9) Communication and Transportation:

The type of transportation used in an area is greatly influenced by weather and climate. Air-conditioned vehicles are popular in tropical climates while such vehicles are not popular in polar regions.

10) Occupation:

The climate of a place does determine the occupation of the people in that particular region.

11) Cultural Activities:

The climate of a place can also affect the cultural activities of some people, e.g. Winter or Summer sports.

Assignment: 1. Briefly describe the Greek classification of climate
2 mention 3 characteristics of tropical rainy climate

Week: six

Topic: climatic classification

Classification of Climate

Classification of climate is a conscious attempt at grouping different climates with similar characteristics together, this is because climate varies from place to place.

The common systems of classification of climate include the Greek and Koppen's classification.

Greek System of Classification of Climate

This is one of the earliest, and most simple and rough climatic classification. This system was put forward by the ancient Greek people. The basis for this classification is Temperature. This system divides the world into three climatic zones:

i) Torrid Zone

ii) Temperate Zone

iii) Frigid Zone

i) Torrid Zone — this zone is found within the tropics

This zone is very hot.

It has high temperature throughout the year.

ii) Temperate Zone

It has mild or moderate temperature.

It has seasonal contrasts in temperature.

This zone is found between the Torrid and Frigid zone.

iii) Frigid Zone

This zone is generally cold, with low temperature throughout the year

This zone is found in Polar Regions (Arctic and Antarctic) regions.

It has a lot of ice -caps in the greater larger part of the year

Koppen's System of Classification of Climate

The basis for Koppen's classification is temperature and rainfall (precipitation). He identified five major climatic groups which correspond with a letter, as follows:

A — Tropical Rainy Climate

This is a moist and hot climate.

It has an average monthly temperature above 18°C.

It has heavy rainfall throughout the year which exceeds annual evaporation.

B — Dry Climates

The annual evaporation exceeds annual rainfall.

It lacks surplus water and is arid.

No river flows permanently in this region.

C — Warm Temperate Rainy Climate (Humid mesothermal climate)

The average temperature of the coldest month is below 18°C but above -3°C.

The average temperature of the warmest month is over 10°C.

It has both Summer and Winter seasons.

D — Cold Snowy Forest Climate (snow and cold climate)

The average temperature of the coldest month is below -3°C.

The average temperature of the warmest month is above 10°C.

It has frozen ground and snow cover for several months duration.

E — Polar Climates (Ice Climate)

The average temperature of the warmest month is below 10°C

There is lack of warm season

Precipitation is in the form of snow

These five major climatic groups are further sub-divided into different subgroups for each of the climate, thus:

A has Af, Am and Aw subgroups

B has BS and BW subgroups

C has Cw, Cs and Cf subgroups

D has Df and Dw subgroups

E has ET and Ef subgroups

The small letters f,s,w, and m reflect how seasonal precipitation is and they mean the following

F= no dry season generally

s= no dry season in summer

w= no dry season in winter

m= moderate dry season in winter and summer

The capital letters of S and W show the two sub division of dry climate

S = mean Steppe or semi-desert

W = mean desert or arid

In polar climates

T = means Tundra

F = means ice cap

Assignment:

1 Define environmental resources

2 mention 5 types of environmental resources